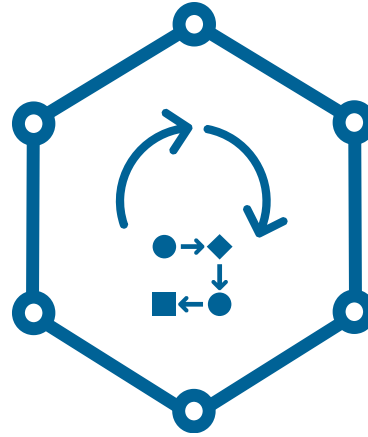




Knowledge Graph Evolution View and Truth Maintenance

Bonus

Emanuel Sallinger



STANFORD

Slides courtesy of Vinay K. Chaudhri
University of Stanford

View Maintenance

- A mechanism from databases to name a query
 - Query is defined with respect to one or more tables (known as **base tables**)
 - If we store the results of the query, the stored data is called **materialized** view
- If the base data changes, the materialized view must be updated
 - Incremental view maintenance

View Maintenance

- A mechanism from databases to name a query
 - Query is defined with respect to one or more tables (known as **base tables**)
 - If we store the results of the query, the stored data is called **materialized** view
- If the base data changes, the materialized view must be updated
 - Incremental view maintenance

Use of view maintenance is not prevalent in current Knowledge Graph engines

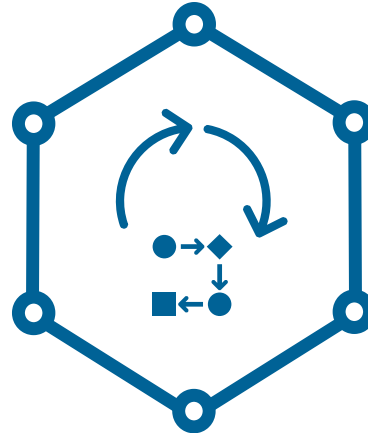
Truth Maintenance

- A mechanism from rule-based systems
 - Tracks how each conclusion was derived
- A popular implementation: Justification based system
 - Each derived conclusion records the fact or rule that was used in derivation
 - Any time that fact or rule updates, the conclusion must be revised

Use of truth maintenance is not prevalent in current Knowledge Graph engines

Summary

- Knowledge Graphs have a life-cycle
 - Must evolve over a period of time
 - Must address both social and technical concerns
- Techniques and algorithms
 - Schema evolution
 - View maintenance
 - Truth maintenance



Knowledge Graph Evolution View and Truth Maintenance

Bonus

Emanuel Sallinger